Quantum Benchmarks

Hardware Runs

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This benchmarking report was generated using Classiq v1.6.0.
For further information, see the Benchmarking directory in classiq-library.

Adder

We benchmark an out-of-place addition by a constant,

$$|x\rangle_n|0\rangle_n \to |x\rangle_n|x + c \bmod 2^n\rangle_n,$$

where c is a fixed integer. In the benchmark, the input variable is prepared in a uniform superposition, and the constant is chosen as

$$c = 2^{\lfloor n/2 \rfloor} - 1.$$

The score is defined as the total measured probability of obtaining the correct arithmetic relation,

$$\text{Score} = \sum_{x,y=0}^{2^n-1} P(x,y)\, \delta_{y,\ x+c \bmod 2^n},$$

where $P(x,y)$ is the measured joint output distribution of the two variables. Equivalently, the score is the total probability mass on outcomes satisfying

$$y = x + c \bmod 2^n.$$

A score of 1 corresponds to perfect agreement with the ideal adder transformation.

adder - 4 qubits (1000 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Depth	2Q Gate Count
Classiq	simulator	1.0	0.0	–	–
Classiq	simulator_statevector	1.0	0.0	104	92
IBM Quantum	ibm_kingston	0.204	395.2	–	–
IonQ	qpu.forte-1	0.877	15.1	–	–

ghz - 3 qubits

Provider	Backend Name	Score	Time Elapsed (min)
Classiq	simulator	0.9972	0.0
Classiq	simulator_statevector	1.0	0.0
IBM Quantum	ibm_kingston	0.4768	379.9
IonQ	qpu.forte-1	0.9609	10.3

grover - 4 qubits

Provider	Backend Name	Score	Time Elapsed (min)
Classiq	simulator	1.0083	0.0
Classiq	simulator_statevector	1.0	0.0
IBM Quantum	ibm_kingston	0.3082	50.9
IonQ	qpu.forte-1	0.6241	10.3

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localization - 3 qubits

Provider	Backend Name	Score	Time Elapsed (min)
Classiq	simulator	0.9154	0.0
Classiq	simulator_statevector	0.9147	0.0

qft - 4 qubits

Provider	Backend Name	Score	Time Elapsed (min)
Classiq	simulator	1.0	0.0
Classiq	simulator_statevector	1.0	0.0
IBM Quantum	ibm_kingston	0.6853	86.6
IonQ	qpu.forte-1	0.9973	15.1

Quantum Volume 2–4 (10 trials)

Provider	Backend	Quantum Volume	Trials	Average Elapsed Time (min)
Classiq	simulator	16	30/30	0.01
Classiq	simulator_statevector	16	30/30	0.01
IBM Quantum	ibm_kingston	8	29/30	439.19
IonQ	qpu.forte-1	8	29/30	12.8

state_preparation - 4 qubits

Provider	Backend Name	Score	Time Elapsed (min)
Classiq	simulator	0.948	0.0
Classiq	simulator_statevector	1.0	0.0
IBM Quantum	ibm_kingston	0.8535	479.3
IonQ	qpu.forte-1	0.8666	10.1

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